Business Problem Statement

A retail company wants to understand customer shopping behavior to improve sales, customer satisfaction, and long-term customer loyalty.

The company wants to analyze customer demographics, product categories, purchase amounts, discount usage, customer reviews, subscription status, shipping preferences, and previous purchases.

By analyzing this data, the company can identify shopping trends and make better decisions about marketing strategies, product recommendations, and customer retention.

Key Business Questions

1. How does purchase revenue vary by gender?
2. Which products have the highest average review ratings?
3. How do purchase amounts differ between Standard and Express shipping?
4. How do subscribers and non-subscribers compare in average spending and total revenue?
5. Which products have the highest discount usage rates?
6. How can customers be segmented based on their previous purchases?
7. Which are the top 3 most purchased products in each category?
8. Are repeat buyers more likely to subscribe?
9. How does revenue vary across different age groups?
10. Do customers who use discounts spend more than the average purchase amount?

Project Deliverables

1. Python: Data cleaning, preprocessing, and exploratory data analysis.

2. SQL (MySQL): Analyze customer shopping data and answer business questions using SQL queries.

3. Power BI: Create an interactive dashboard to visualize customer shopping behavior and key business metrics.

4. Project Report: Document the project workflow, SQL analysis, key findings, and business recommendations.

5. GitHub Repository: Upload the project files, Python code, SQL queries, dashboard, and documentation.

6. Expected Outcome

To identify customer shopping patterns, understand customer preferences, and provide data-driven recommendations to improve sales, customer engagement, and customer loyalty.